

ASSIGNMENT 1

Characterising the Urban Heat Island

This assignment follows directly from the urban heat island tutorial.

Read section 4 before you start. The most important part of focus for the assignment is not producing a map it is for explaining why you made the choices you made.

1 What you have to produce

Two linked analyses of the same urban area, written up in report.

Part A - Day and night UHI from Landsat

- Compute land surface temperature for one clear daytime scene and, where one is available, one night-time scene over your study area.
- Define urban and rural zones from a land cover product, and compute UHI intensity as the difference of their means.
- Produce an LST map, an anomaly map, and a distribution plot comparing the two zones.
- Interpret the difference between the day and the night result.

Pick a city or urban area which has night-time Landsat scene

Night acquisitions are rare and irregular, and many cities have none at all. This is a real constraint so make sure to find a location which has it and fill it out in the shared list.

Part B - Seasonal UHI from MODIS

- Using MODIS land surface temperature, characterise how UHI intensity varies across two or three seasons, defined appropriately for your city's climate.
- Use the same urban and rural zone definitions as Part A, so the seasonal comparison is about season and nothing else.
- Report MODIS UHI for each season.
- Compare your MODIS result against your Landsat result for the overlapping season, and account for any disagreement.
- Feel free to report any other additional inference from your analysis including any other substantive results.

Seasons are yours to define, and you must defend the definition with valid reference

State the months you assigned to each season and cite the climatological basis.

2 Your study area, and where the boundary comes from

You must define your area of interest from a published vector boundary file. You may not draw it by hand in the Code Editor, and you may not use a plain circular buffer as your only definition of the city. An administrative city limit, a built-up extent derived from satellite data, and a metropolitan planning region will each give you a different UHI for the same place.

What you must report about your boundary

- The source, its version or release, and the date you downloaded it.
- A full citation in your reference list, including the URL and access date.
- The coordinate reference system of the file, and what you did about reprojection.
- Which administrative or functional level you selected, and why that level suits the question.
- The area of the resulting polygon, in square kilometres.

Example citation. Global Administrative Areas (2022) *GADM database of Global Administrative Areas, version 4.1*. Available at: <https://gadm.org/data.html> (Accessed: 14 March 2026).

Adapt to whichever referencing style your module requires.

3 Data products

These are the collections you are expected to use. Others are also acceptable with justification and reference.

PURPOSE	EARTH ENGINE ASSET
Landsat day LST	LANDSAT/LC08/C02/T1_L2 (and T2_L2)
Landsat night	LANDSAT/LC08/C02/T1 (and T2)
MODIS seasonal	MODIS/061/MOD11A2 (Terra) MODIS/061/MYD11A2 (Aqua)
Land cover	ESA/WorldCover/v200/2021 (or any other LULC data available for your AOI)
Cloud screening	QA_PIXEL, both Landsat levels

A choice you must make and defend: Terra or Aqua

Terra crosses at roughly 10:30 and 22:30 local time. Aqua crosses at roughly 13:30 and 01:30. Aqua therefore sits much closer to the daily maximum and the pre-dawn minimum, which is usually what a heat island study wants. Terra's record is longer, and its orbit has drifted over the mission, so its overpass time is not constant across years.

Pick one, or use both. Either is defensible. Make sure to mention.

4 Justifying your decisions (read this carefully)

Majority of the marks on this assignment are for justification and interpretation. That is more than for producing the maps.

At every point below, your report must state what you chose, what the alternatives were, and why you chose as you did.

DECISION POINT	THE QUESTION YOUR REPORT MUST ANSWER
Study area boundary	Administrative limit, or built-up extent? What would have changed if you had chosen the other?
Urban definition	Which land cover classes count as urban, and what did you do with the classes that are neither clearly urban nor clearly rural?
Rural reference	What did you exclude, and why? Did you exclude water? How far from the city edge does your rural zone start?
Erosion radii	How much did you erode each mask, and how much did your UHI change when you varied it? Report the range, not just one number.
Scene selection	Which scene, on what date, with what cloud cover and what valid-pixel fraction over your AOI? Why that one and not another?
Cloud masking	Which QA bits did you mask and why? What fraction of your AOI survived?
Emissivity	Which method, and what did applying it change? If you reported brightness temperature rather than LST, say so plainly.
Season definition	Which months, on what climatological basis, and cited from where?
Terra or Aqua	Which platform, and what does its overpass time mean for what your seasonal numbers represent?
Statistics scale	At what spatial scale did you compute the zonal means, and how many pixels went into each?

The sensitivity test is compulsory

For at least one decision make the parameter across a sensible range and report how much the UHI intensity moves. You must interpret the result carefully.

5 Submission checklist

- Report as a single PDF, named RollNo._UHI.pdf.
- Earth Engine script shared with view access, and the link pasted in the appendix.
- Your script runs top to bottom without editing, and its parameters match what the report says you used.
- Every dataset and boundary file cited in the reference list.
- AI-use declaration completed - see below.

6 Use of AI tools

The tutorial script is provided to you and you may adapt it freely. You do not need to cite it, but you must not present it as your own original design. Everything beyond it must be your own work.

AI assistants may be used for explaining concepts, debugging code you have written, and improving the clarity of your prose. They may not be used to generate your justifications, or your interpretation. Those are precisely what is being assessed.

Declaration required

Include a short paragraph at the end of your report stating which AI tools you used, if any, and for what. An undeclared one, if detected, is treated as an academic offence. If you used no AI tools, say that.

A practical warning. AI assistants are confidently wrong about Earth Engine specifics: scale factors, QA bit positions, band names, and which functions exist. The tutorial deck covers several cases where a plausible-looking answer produces a map that is quietly incorrect. Verify anything an assistant tells you against the Earth Engine catalogue page or the USGS product guide, and cite the documentation rather than the assistant.

Ask questions early. A boundary-file problem raised on day two is a five-minute fix..